

A 1.2GHz 12.77GB/s/mm² 3D Two-DRAM-One-Logic Process-Near-Memory Chip for Edge LLM Applications

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<< Results | < Previous | Next >

Yue Cao ; Jinghao Jiang ; Haijun Jiang ; Qian Zhang ; Xuanzhi Liu ; Jinhui Cheng ...

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Abstract:
A high-bandwidth-density (12.77GB/s/mm²) high-memory-density (99.4Mb/mm²) low-energy-consumption $\$(0.67 \text{ } \backslash \text{text{pJ}}) / \dots$ [Show More](#)

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Contents

Upon the rapid development of generative AI, there is an urgent need to embed LLM inference support into client devices (PCs, mobiles) to improve response time, data privacy, and customization. However, as shown in [Fig. 30.7.1](#), challenges remain for edge-LLM applications. Compared to cloud deployments, edge tasks typically use much smaller batch sizes, which significantly reduces data reuse during the generation stage of LLM inference and makes the task more memory-intensive [1]. Meanwhile, existing client products are limited by either computing power (CPU) or memory bandwidth (GPU), and GPU energy consumption is prohibitive for battery-powered devices such as phones. Therefore, supporting offline edge LLM inference requires an LLM-optimized design that delivers sufficient computing power and memory bandwidth, high energy efficiency, and easy integration with existing edge platforms.

To mitigate the memory bottleneck, several processing-near-memory (PNM) architectures have been proposed [2]– [5]. However, most prior works integrate compute directly within the DRAM die, constraining compute capability due to DRAM process limitations. In parallel, recent advances in 3D HB + mini-TSV integration enable high-bandwidth low-energy memory access for 3D near-memory designs, motivating this work. Our design stacks two DRAM dies with one logic die, providing 1GB on-chip DRAM capacity and a memory density of 99.4Mb/mm². Decoupling logic from DRAM allows implementing compute logic in standard CMOS and seamlessly leveraging future improvements via advances in process nodes. The 3D-stacked DRAM delivers GB-scale on-chip memory for LLM deployment, greatly reducing off-chip traffic, while the high-density HB + mini-TSV interconnect enables near-in-die interconnects that offer low-latency, low-energy memory access, achieving a bandwidth density of 12.77GB/s/mm² to alleviate the LLM



memory bottleneck. The design also supports a DDR4-protocol-compatible external interface for flexible system integration; when PNM is disabled, the chip operates as conventional host DRAM.

[Figure 30.7.2](#) illustrates the overall architecture of the presented 3D NMC design. The presented design comprises eight 1 Gb compute blocks, denoted as bankgroup (BG) blocks. Each BG block consists of two 512 Mb DRAM blocks on the DRAM_N and DRAM_F dies, and a compute module on the logic die. Each DRAM block contains 8 banks and each bank contains 8 segments. Within each segment, 16 subarrays are arranged such that two adjacent subarrays share a 512-SA stage to generate local DQ (LDQ) signals. By controlling MDQ_sel, target column data are selected onto MDQ lines and then driven into a secondstage SA for faster reads. Each segment produces a 16b RDL output, so one BG block provides a 2048b bus ($= 16 \times 8 \times 8 \times 2$). These RDLs connect to the logic die via 3D paths: signals from DRAM_N use direct HB connections, while those from DRAM_F traverse HB-TSV-HB chains to pass through DRAM_N. For TSV reliability, two HB connections accompany each TSV. The RDLs from the DRAM dies connect to a custom 3D DRAM PHY. Because the 3D stack yields a much wider bus, the pin data rate required for equivalent bandwidth is substantially lower; consequently, the 3D DRAM PHY is simpler than DDR/HBM PHYs and primarily serves as a data synchronizer and level shifter, reducing design complexity and power consumption.

Each compute module integrates two accelerator (ACC) units that perform near-memory computation and access DRAM through the memory controller. The general-purpose system host accesses DRAM and manages the ACCs via a DDR-compatible interface. Two additional DDR4-formatted commands, T-ACC and Act-Reg, are introduced to trigger target ACCs and set configuration registers. When the host issues a command, the DDR interface and decoder accept and decode it, and then assert the corresponding control signals to the target module (ACC or memory controller). This scheme maintains full host compatibility while providing a lightweight command extension mechanism for near-memory control.

[Figure 30.7.3](#) shows the 3D DRAM datapath, including ACC-DRAM and system-memory/DRAM paths. Our RISC-V-based ACC includes a core that supports common RISC-V instructions. L1 instruction and data caches reduce access latency. An SRAM-based system memory is connected to the ACC via the memory bus for local data storage. We designed a custom FPU to accelerate critical floating-point operations for LLM workloads (fadd, fsub, fmul, fdiv); other floating-point instructions, such as fsqrt, are also supported by the core. As LLM workloads involve extensive consecutive loads and stores, a DMA module manages bulk transfers to relieve the core of data-movement tasks. The ACC can issue direct DRAM access commands or trigger DMA transfers between system memory and 3D DRAM space. The memory controller decodes access requests from the system bus and forwards them to the command scheduler, which generates DRAM commands. The scheduler FSM is also shown in [Fig. 30.7.3](#). We employ an open-page policy and add a 2048b buffer to hold the most recently accessed data. On a new request, if buf_hit is asserted, the scheduler serves data from the buffer; otherwise, it starts a DRAM request. For a row hit, the read/write command is issued directly; on a row miss, precharge and activate commands are scheduled before read/write. On DRAM reads, the returned data updates the buffer. Scheduled commands flow to a signal generator to form actual DRAM control signals, and the corresponding timing waveform is presented in [Fig. 30.7.3](#). A host programmable config module enforces timing and refresh scheduling. A refresh control issues refresh requests per the configuration and forwards them to the scheduler. Generated DRAM signals and external host signals are selected by a switch according to the current mode and remapped to DRAM addresses. The 3D DRAM PHY comprises four 256 Mb PHY submodules to service the 1 Gb DRAM space.

[Figure 30.7.3](#) also illustrates the GEMM dataflow. GEMM is the dominant kernel in LLM computation, thus our evaluation focuses on this task. Cross-BG or cross-chip data accesses require host intervention and should be avoided whenever possible. For example, in a full-chip assignment for an $M \times N \times K$ GEMM ($M \times N$ input multiplied by $N \times K$ weight), M and K are each partitioned by four, and the resulting submatrices are assigned to the 16 ACCs across the 8 BGs (e.g., $IOW0 \rightarrow BGO/ACCO$, $IOW1 \rightarrow BGO/ACC1$, $IOW2 \rightarrow BG1/ACCO$,...). Each accelerator computes a submatrix of the final output, which the host can directly concatenate without additional accumulation. When scaling to multiple NMC chips, M and K can be further partitioned. Within each accelerator, all weight data are stored consecutively and read via DMA, while each input row is evenly distributed across all banks accessed sequentially, enabling complete input reads through all-bank parallelism. When an input row is ready, it will be sent to the digital logic with the corresponding weight data for MAC and potentially activation function (AF) computation. Normalized GEMM performance results show that adding DMA and the 2048b buffer yields substantial improvement: on average, throughput increases by 53.8% relative to a baseline without DMA or buffer, demonstrating the benefits of local buffering plus DMA-assisted data movement. Furthermore, in some specific scenarios,

parameters like weights can be uploaded to 3D DRAM only once after powering on, which could greatly reduce the off-chip access amount.

Figure 30.7.4 presents the measurement system and prototype board. The setup uses one host FPGA and two test PNM chips to verify scalability. A host CPU in an FPGA (serves as the general-purpose CPU system in application scenarios) accesses the test chips via the host memory controller and a DDR4-compatible interface. Control signals (commands and addresses) are broadcast to both chips. The 16b DDR4 data bus is evenly split and connected to the two test chips. Inside each test chip, control and data signals are broadcast to the 8 BGs, and the target BGs activate according to decoded information. A high-speed IO chip is used for ACC program loading and debug transfer, providing program flexibility.

Figure 30.7.4 also shows DRAM measurement results. The DRAM shmoo indicates operation at 526 Mbps at 1.2 V, corresponding to 12.77GB/s/mm² bandwidth density, which exceeds GDDR6 by 8.63× and HBM3E by 1.22×. DRAM retention testing (96 ms at 98 °C) failed on <0.05% of wafer chips, indicating good DRAM quality. The results of 3D integration yield verification, and the corresponding testing circuit is also presented in Figure 30.7.4. Only 0.58% additional failure rate is detected after 3D integration, proving that the integration technology maintains a good yield rate. The 3D integration architecture enables 0.67pJ/b memory energy efficiency, which is 9% of GDDR6 and 28% of HBM3E.

As shown in Fig. 30.7.5, the 3D-stacked chip integrates high-density HB and mini-TSV interconnects with HB pitch = 3μm and TSV pitch = 5μm. Figure 30.7.5 reports voltagefrequency scaling, logic-die area breakdown, and chip power breakdown. It also presents memory-access latency and GEMM execution results. In an 8192-access consecutive test, we measure 3.21 ns write and 3.24 ns read latency, reductions of 90% and 93% versus a scaled LPDDR4 system (system frequency scaled to 1.2 GHz); for random access, we measure 45.12 ns write and 50.01 ns read latency, reductions of 37% and 44%. These results show that the 3D near-memory architecture substantially reduces memory latency, benefiting latency-sensitive edge LLM workloads. Compared to a 2.1 GHz Intel i7-13700F, our design reduces execution time for a 128 × 128 × 128 GEMM by 86%. When scaling our work to an 8-chip DIMM system with 8GB DRAM capacity and 128 ACCs, the reduction can be further improved to 98%, illustrating practical gains for common inference kernels.

Figure 30.7.6 compares our design with SOTA PNM designs. Unlike prior works, our design supports a DDR4-compatible interface for general-purpose integration. The twodie stacked DRAM with 3D integration achieves 12.77GB/s/mm² bandwidth density, 99.4Mb/mm² memory density, and 0.67pJ/b energy efficiency, while a separate 28 nm CMOS logic die enables 1.2 GHz compute frequency, which provides suitable computing power for client LLMs. The design FoM is 7.33× higher than a prior 3D PNM design and 5.13× higher than an active-interposer design [7]. By consolidating the logic and DRAM dies through 3D integration, this work leverages PNM-DRAM advancements without compromising the compute capability provided by standard CMOS logic, achieving outstanding bandwidth density and logic frequency compared with existing PNM designs. Moreover, this industrial-standard fabricated prototype demonstrates the feasibility and commercial potential of 3D PNM solutions for edge LLM deployment. The die micrographs and key specifications are shown in Fig. 30.7.7.



Figure 30.7.1:
Client-side LLM application challenges and the presented 3D integrated near-DRAM computing design.



Figure 30.7.2:
Overall design architecture and the extended DDR4 command list.



Figure 30.7.3:
Datapath and access timing for 3D DRAM, and dataflow and performance test for GEMM.



Figure 30.7.4:
Measurement system architecture and 3D DRAM measurement results.



Figure 30.7.5:
3D integration details, area and power breakdown, and design measurements.



Figure 30.7.6:
Comparison with prior works.



Figure 30.7.7:
Die micrographs and design specifications.

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